MATH 9102

ASSUMPTIONS FOR LINEAR REGRESSION

Sources used in creation of this lecture:
Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using IBM SPSS, Andy Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

Fundamentals

Multiple linear regression requires at least two independent variables (predictors)

- Can be nominal, ordinal, or interval/ratio level variables.

A rule of thumb for the sample size is that regression analysis requires at least 20 cases per independent variable in the analysis.

Example – MATH9102-W9-Assumptions.QMD

Using Pallant's Survey dataset

Model1

- Investigate whether perception of control can be considered to predict perceived stress levels controlling for a respondent's likelihood to respond in a socially desirable manner
- Outcome variable: Perceived Stress
- Predictors: Perception of Control and Social Desirability
- `model1=lm(sdata$tpstress~sdata$tpcoiss+sdata$tmarlow)`

Model2

- Investigate whether perception of control and optimism can be considered to predict perceived stress levels controlling for a respondent's likelihood to respond in a socially desirable manner
- Outcome variable: Perceived Stress
- Predictors: Perception of Control, Optimism and Social Desirability
- `model2=lm(sdata$tpstress~sdata$tpcoiss+sdata$toptim+sdata$tmarlow)`

Model3

- Investigate whether perception of control and optimism can be considered to predict perceived stress levels controlling for a respondent's likelihood to respond in a socially desirable manner and whether a differential effect exists for gender
- Outcome variable: Perceived Stress
- Predictors: Perception of Control, Optimism and Social Desirability
- `model2=lm(sdata$tpstress~sdata$tpcoiss+sdata$toptim+sdata$tmarlow+sdata$gender)`

Straightforward Assumptions

Variable Type:
- Outcome must be continuous
  - **Note: does not have to be normal but will make your life easier if it is**
- Predictors can be continuous or dichotomous (binary nominal).

Independence of observations:
- All values of the outcome should come from a different observations.

Linear relationship:
- There must be a **linear relationship** between the outcome (dependent) variable and the independent (predictor) variables.
- Scatterplots can show whether there is a linear or curvilinear relationship.
- Correlation and difference tests can provide evidence of statistical significance.

Note: We have established all of these previously.

Step 1: Check Non-Zero Variance

Predictors and Outcome must not have zero variance.

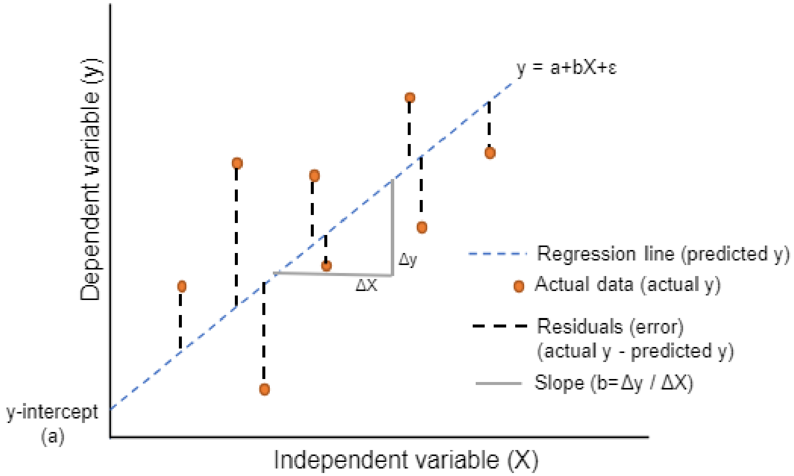
- Check your descriptive statistics
  - pastecs::stat.desc(sdata$tpstress, basic=F)
  - pastecs::stat.desc(sdata$tpcoiss, basic=F)
  - pastecs::stat.desc(sdata$tmarlow, basic=F)

- How to report e.g. for Model 1:

"The data met the assumption of non-zero variances (Perceived Stress = 34.78, Perception of Control = 146.50, Social Desirability = 4.02). "

Step 2: Residual Analysis

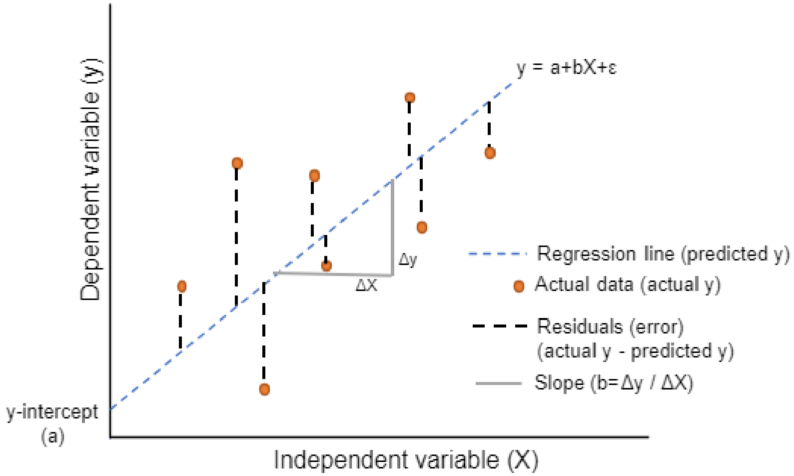
Residuals



Coefficients and statistical significance are calculated on the assumption that a straight line is a good model of the relationship

How well this line serves as a model of this relationship can be checked by looking at how well the observed data sit in relation to predicted values along this line

- Measured by the vertical distance from the prediction to the actual observation – the **residual**



Residual Analysis

Regression coefficients are calculated so that the resulting line has the lowest possible accumulation of residuals, minimising the overall distance between the observation and the predictions

You need to conduct a residual analysis to identify potential issues.

Residual Analysis

Outliers:

- An observation that has a large residual i.e. the observed value for the point is very different from that predicted by the regression model.

MLR can be greatly influenced by outliers

- They ‘pull’ the equation away from the general pattern

Residual Analysis

Leverage points

- Observations with predictor (x) values that are substantially distant from the mean of the predictors.
- Because they occupy extreme positions in the predictor space, they have the potential to exert a disproportionate effect on the fitted regression line by “pulling” it toward themselves.
- Leverage is quantified using the **hat matrix (H)**, where higher leverage values indicate greater potential influence, though not all high-leverage points necessarily distort the model.

Influential observations

- Cases (rows) that meaningfully alter the estimated regression coefficients or overall model fit when included or excluded from the analysis.
- They combine both leverage (unusual x-values) and large residuals (unusual y-values), thereby having a strong effect on the slope or direction of the regression line.
- Influence is commonly assessed using **Cook’s Distance**, which evaluates the impact of each case on the model’s parameter estimates.

Residual Analysis Core Assumptions- Must be tested post hoc (after the model is created)

Normality of residuals

- Multiple regression assumes that the **residuals are normally distributed**.
- If your outcome variable is normally distributed, then this will be much easier to achieve.

No influential outliers or leverage points

- Outliers or influential data points can disproportionately affect the regression results. Leverage and Cook's Distance can be used to identify potential outliers or influential points.

Homoscedasticity of residuals

- The **variance of error terms** should be similar across the values of the independent variables (predictors)
- A plot of standardized residuals versus predicted values can show whether points are equally distributed across all values of the independent variables.

No Multicollinearity

- Multiple regression assumes that the **independent variables (predictors) are not highly correlated** with each other.
- This assumption is tested using Variance Inflation Factor (VIF) values.

Residual Analysis: Step 2 a: Conduct a Normality Test

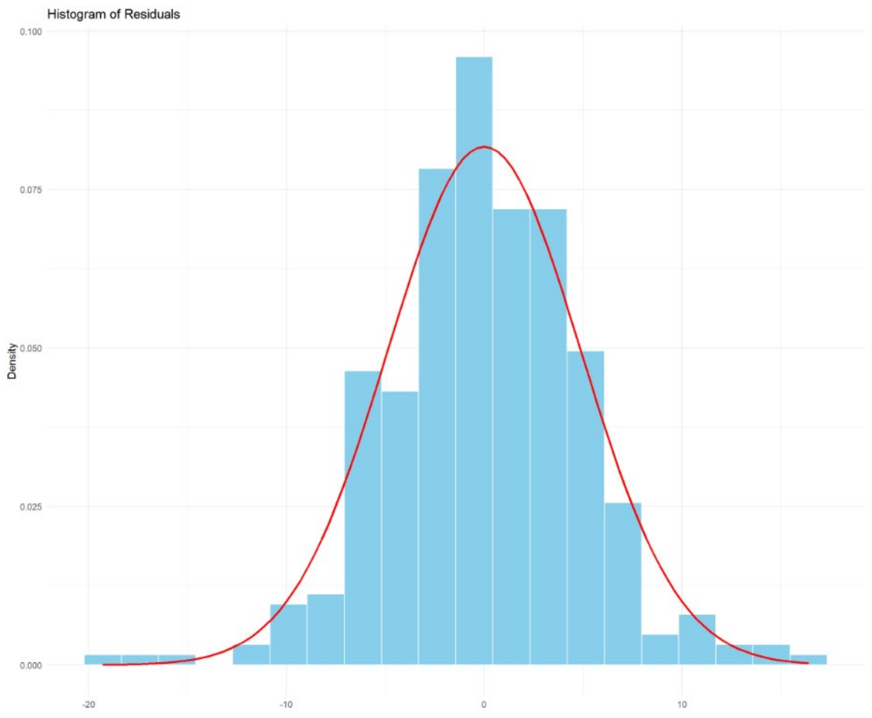
Standardized residuals should follow a normal distribution with a mean of 0

Since the residuals measure where the points fall in relation to the regression line, a normal distribution of residuals indicates that the same number of points fall above and below the line

A residual of 0 means a point is on the line, a mean of 0 means that the line is in the middle of the points

```
performance::check_normality(model1)
```

Step 2b: Visualize residual distribution



First check the histogram of the residuals:
Blue shaded area is the residual distribution

Red line is the normal curve for comparison

Gives us an indication that the non-normality is low

Q-Q Plot

Q–Q Plot (Quantile–Quantile):

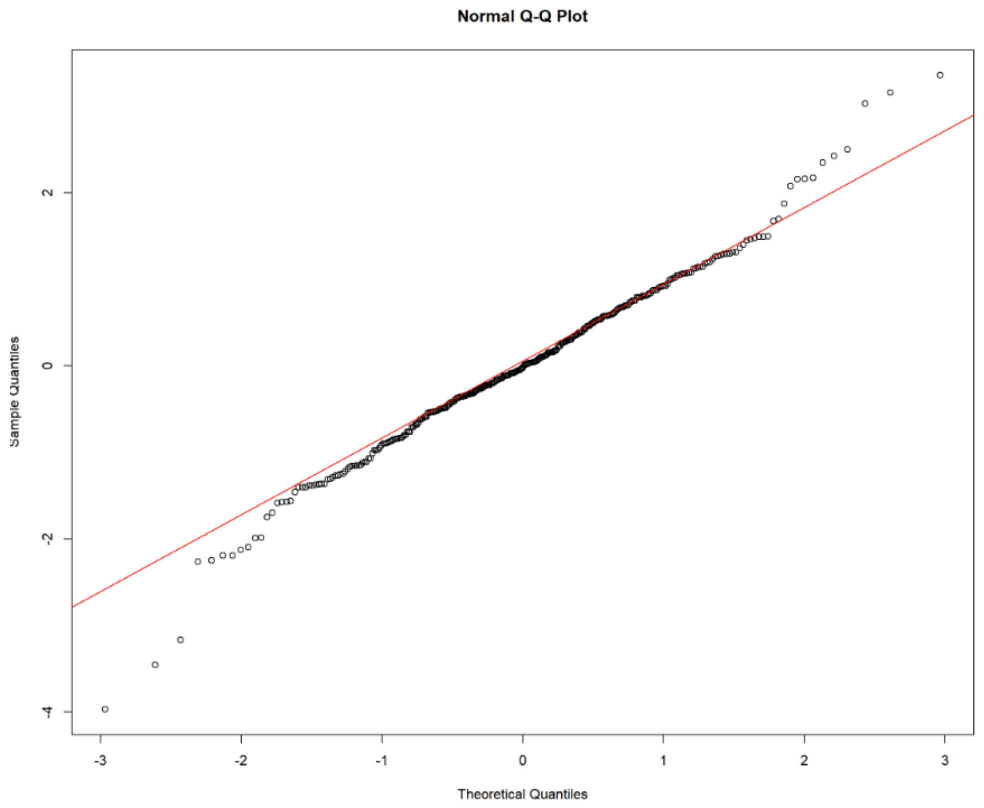
- Compares the *quantiles* of your sample data (e.g., residuals) to the quantiles of a theoretical normal distribution.
- If residuals are normally distributed, the points fall roughly along a **straight diagonal line**.
- Deviations from the line indicate skewness or non-normality, especially in the tails.

Step 2b: Visualize residual distribution

Check the Q Q plot:

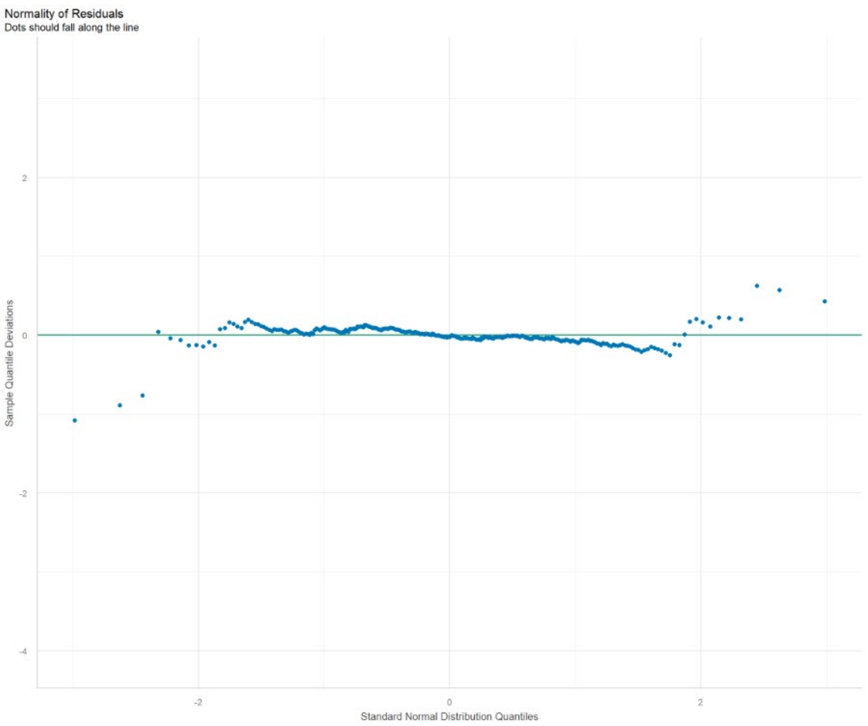
Red diagonal is the reference line.

There are some areas of concern



Step 2b: Visualize residual distribution

A *transformed version* of a Q–Q plot displayed horizontally



Residual Analysis: Step 2c: Check outliers or influential points

Influential outliers are data points that have a **disproportionate impact** on the regression model's estimates.
They are typically identified using **Cook's Distance**, which combines both:

- how far an observation's predictors are from the mean (leverage), and
- how far its residual is from the fitted value (error).

Calculate **Cook's distance**

- Quantifies the effect of removing a data point on the estimated regression coefficients.
- There is one Cook's D value for each observation used to fit the model
- A value significantly greater than zero suggests that the data point is influential and may be an outlierthe model.

```
performance::check_outliers(model1)
```

$$D_i = \frac{(r_i^2/p) \cdot h_{ii}}{(1-h_{ii})^2}$$

Where:

- r_i is the residual for the i-th observation.
- h_{ii} is the leverage for the i-th observation.
- p is the number of parameters in the model.

Residual Analysis: Step 2c: Check outliers or influential points

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Where:

- r_i is the residual for the i-th observation.
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Steps to check:

Calculate Cook's Distance for each case (usually available via diagnostic output in R).

Examine values:
- **Cook's D > 1** → generally considered highly influential.
- **Cook's D > 4 × mean(Cook's D)** → a **more conservative threshold**, useful for larger datasets where most values are small.

Inspect any cases exceeding these thresholds — they may warrant further investigation or sensitivity testing (e.g., re-running the model without them).

Interpretation:
- High Cook's Distance values suggest observations that significantly alter regression coefficients if removed.
- These should be checked for data entry errors, unusual characteristics, or legitimate but high-impact cases.

Cook's Distance Our Example

No issues identified

Step 2c: Check outliers or influential points

```
# Check for outliers or influential points

performance::check_outliers(model3)
```

```
OK: No outliers detected.
- Based on the following method and threshold: cook (0.9).
- For variable: (Whole model)
```

Residual Analysis: Step 2d: Examine the Residuals

Look at the Minimum and Maximum values of Std. Residual (Standardised Residual) subheading.

- For small datasets (80 or less)
  - We would expect 95% to fall within +1.96 and -1.96 if our p value cut-off is 0.05 (5%)
- For large datasets (80 or more)
  - If the minimum value is equal or below -3.29, or the maximum value is equal or above 3.29 then you have outliers.
- We need to investigate whether these outliers are having a significant influence on our results.

- For our example:
  - Max standardized residual: 3.477873
  - Min standardized residual: -4.09056
  - So there is a need for further investigation

Residual Analysis: Step 2d: Examine the residuals

Identify potential outliers in the dataset based on the studentized residuals of the model.

```
car::outlierTest(model1)
```

This will calculate a Bonferonni p-value for most extreme observations

Observations with a p value lower than the threshold we are working to (e.g. 0.05) are consider outliers.

In our case for model 1, there is one outlier – observation/row/case number 39

```
    rstudent unadjusted p-value Bonferroni p
39 -4.060219         6.1324e-05     0.020421
```

Studentized Residuals

Studentized Residuals:

- Residuals divided by an estimate of their own standard deviation.
- This adjustment accounts for varying residual variance and **leverage**, making the residuals more comparable across observations.
- They are used in diagnostic plots because they provide a standardized measure of how far each point lies from the regression line.

Residuals Concerns for MLR – What do you do with outliers?

Sometimes it is reasonable to delete the outliers

Sometimes you need to retain the outlier

- it may be your most important observation

Sometimes it is a good idea to do your analysis twice – once with and once without the outlier to see how much influence it is having on the model

If you remove outliers, do repeat the residual analysis again, you may have introduced new outliers

How to report MLR

Outliers

- Look at the Minimum and Maximum values
- If the minimum value is equal or below -3.29, or the maximum value is equal or above 3.29 then you have outliers.
- Look at Cook's distance for values greater than one
- Identify potential outliers in the dataset based on the studentized residuals of the model.
- Note: If you remove outliers, do repeat this residual analysis again, you may have introduced new outliers

Reporting this (a general example):

"An analysis of standard residuals was carried out on the data to identify any outliers, which indicated two cases for concern which were deleted."

How to conduct and report MLR

If no outliers found report the min and max e.g.

An analysis of standard residuals was carried out, which showed that the data contained no outliers (Std. Residual Min = -1.90, Std. Residual Max = 1.70).”

Step 3: Check homoscedasticity (equal variance)

Residuals should have constant variance across all levels of the independent variables

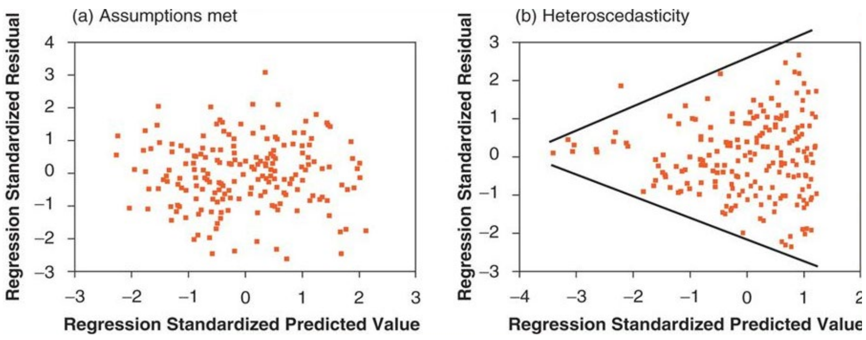
Heteroscedasticity (non-constant variance) can lead to inefficient estimates.

This can be checked using a plot of residuals versus fitted values, or the Breusch-Pagan test.

```
performance::check_heteroscedasticity(model1
)
```

Why are we concerned?
Bias
Increased possibility of Type I error if heteroscedasticity

Homoscedasticity



How to check?

- Scatterplot: plot the residuals against the predicted values of the dependent variable.
- If the residuals are randomly scattered around the horizontal line at zero without forming any patterns or trends, then the assumption of homoscedasticity is likely to hold.
- However, if the residuals show a funnel-shaped or cone-shaped pattern, with the spread increasing or decreasing as the predicted values increase, then the assumption of homoscedasticity is violated

Homocedasticity

Diagnostic Plots (see Step 6 Summary Plots)

- A scatterplot of the prediction errors (residuals) against the predicted values is used to see if the predictions can be improved by fixing problems in your data
- The residuals, $Y - [a + b1X1 + b2X2 + \cdots + bkXk]$, are plotted on the vertical axis
- The predicted values, $a + b1X1 + b2X2 + \cdots + bkXk$, go on the horizontal axis.
- You should only try to fix a problem if it is clear and extreme.
- Do not intervene unless the diagnostic plot shows you a clear and definite problem

Homocedasticity

Tilted linear relationship with outlier(s)

- The outlier(s) have distorted the regression coefficients and the predictions.
- The predictions for the well-behaved portion of the data can be improved if you feel that the outliers can be controlled (perhaps by transformation) or omitted.

Curved relationship, typically U-shaped or inverted U-shaped

- There is a nonlinear relationship in the data.
- Your predictions can be improved by either transforming, including an extra variable, or using nonlinear regression.

Unequal variability

- Your prediction equation has been inefficiently estimated.
- Too much importance has been given to the less reliable portion of the data and too little to the most reliable part.
- This may be controlled by transforming Y (perhaps along with some of the X variables as well).

Step 4: Check Mult-Collinearity

Occurs when two or more independent variables contain strongly redundant information

If variables are collinear then it means there is not enough distinct information in these variables for MLR to operate – they are essentially measuring the same thing

If we conduct MLR with collinear variables then the model will produce invalid results

Before conducting MLR

- Need to check for collinearity by examining a correlation matrix that compares your independent variables with each other.
- A correlation coefficient above 0.8 suggests collinearity ***might*** be present
    - If it is you need to analyse your variables in separate regression equations and then examine how they operate together

Step 4: Check Mult-Collinearity – Post hoc

Calculate the **VIF (Variance Inflation Factor)**

- A variance inflation factor (VIF) quantifies how much the variance is inflated.

Tolerance is the reciprocal of VIF.

- We are interested in Tolerance and VIF (variance Inflation Factor)
- If the VIF > 5 and Tolerance < 0.2 then you have concerns over multicollinearity.

- `performance::check_collinearity(model1)`

Step 4: Check Mult-Collinearity

What can we do if there is a problem?

- Delete all but one of the collinear variables from the model
  - The one remaining is a proxy for the underlying concept
- Combine them into an index mathematically (e.g. multiplying, adding etc.) – must make sense theoretically
- Estimate a latent variable using principal component analysis or factor analysis

How to report

Collinearity

- Coefficients table
- We are interested in Tolerance and VIF (variance Inflation Factor)
- If the VIF value is greater than 5 or the Tolerance is less than 0.2, then you have concerns over multicollinearity.

Otherwise, if your data has met the assumption of collinearity and can be written up something like this:
Reporting this:
*"Tests to see if the data met the assumption of collinearity indicated that multicollinearity was not a concern (Gender, Tolerance = .99, VIF = 1.00; Standard Reading Test, Tolerance = .45 VIF = 2.23, Interaction effect Standard Reading Test * Gender Tolerance=.45, VIF=2.23)."*

Step 5: Report - Model 1

The data met the assumption of non-zero variances (Perceived Stress = 34.78, Perception of Control = 146.50, Social Desirability = 4.02). Examination of diagnostic plots indicated that the residuals were approximately normally distributed, with a density plot resembling normality, although the formal test suggested a slight deviation ($p = 0.003$). The standardized residuals ranged from –3.95 to 3.36, with one observation exceeding ±3 standard deviations. The Bonferroni-adjusted outlier test ($p = 0.020$) indicated that this single case may be a mild outlier, but its influence was not substantial, as Cook's distance remained below the conservative threshold of 1.0. The residuals-versus-fitted values plot displayed a generally random scatter around the zero line, with no evidence of heteroscedasticity ($p = 0.101$), supporting the assumption of constant variance. Collinearity statistics showed low intercorrelations between predictors (Perception of Control VIF = 1.10, Tolerance = .91; Social Desirability VIF = 1.10, Tolerance = .91), confirming that multicollinearity was not a concern. Overall, the model met the key assumptions of multiple linear regression—linearity, independence, homoscedasticity, and low multicollinearity—with only a minor departure from normality and a single non-influential outlier. The model can therefore be considered to provide a reliable estimate of the relationship between perceived stress, perception of control, and social desirability.

Diagnostic Plots (see Step 6 Summary Plots)

LINEARITY (TOP-LEFT)

Purpose: Checks whether the relationship between predictors and the outcome variable is approximately linear.

What to look for:

- The **green reference line** should be **flat and horizontal**.
- The residuals (blue dots) should be **randomly scattered** around the zero line.

Interpretation:

- A flat pattern → linearity assumption met.
- A curve or wave pattern → possible **non-linear relationship** between predictors and the dependent variable.

HOMOGENEITY OF VARIANCE (TOP-RIGHT)

Purpose: Tests homoscedasticity — equal variance of residuals across fitted values.

What to look for:

- The green line should again be flat and horizontal.
- The spread of points should be roughly **even** across the range of fitted values.

Interpretation:

- Constant spread → assumption met.

Diagnostic Plots (see Step 6 Summary Plots)

INFLUENTIAL OBSERVATIONS (MIDDLE-LEFT)

Purpose: Detects outliers and leverage points that could strongly influence the regression line.

What to look for:

- Points should fall **within the dashed Cook's Distance contour lines**.
- Labelled points (e.g., 39, 108, 282) indicate cases with higher influence or leverage.

Interpretation:

- Points inside the contours → not problematic.
- Points outside → potentially **influential**; consider inspecting or testing them further.

COLLINEARITY (MIDDLE-RIGHT)

Purpose: Checks for multicollinearity among predictors.

What to look for:

- Bars represent the **Variance Inflation Factor (VIF)** for each predictor.
- The green region (<5) = acceptable; the red zone (>5) = potential concern.

Interpretation:

- Low VIF (≤5) → predictors are sufficiently independent.
- High VIF (>5) → multicollinearity may be inflating standard errors.

Diagnostic Plots (see Step 6 Summary Plots)

NORMALITY OF RESIDUALS – Q–Q STYLE (BOTTOM-LEFT)

Purpose: Evaluates whether the distribution of residuals approximates a normal distribution.

What to look for:

- Dots should lie close to the **horizontal line at zero** (this is a *Q–Q deviation plot* rather than the classical diagonal Q–Q plot).

Interpretation:

- Dots close to the line → residuals roughly normal.
- Curved or S-shaped patterns → **non-normality** (e.g., skewed or heavy-tailed residuals).

NORMALITY OF RESIDUALS – DENSITY PLOT (BOTTOM-RIGHT)

Purpose: Another check of residual normality, shown as a histogram or smoothed density curve.

What to look for:

- The residual distribution (grey area) should align with the **green normal curve**.
- Centred on zero with symmetric tails.

Interpretation:

- Close match → normality assumption supported.
- Skewed or lopsided shape → possible **departure from normality**.

Conducting MLR

Concerns for MLR - Causality

Need to take care about asserting causality on the basis of your analysis

Causality

- Association
- Time order
- Non-spuriousness

Regression can reveal associations, but they do not document time order

By including other factors, we can control for spurious effects

- But there is always the possibility that there is an untested spurious factor

How to conduct an MLR

Research and develop a theory

Identify your variables and scales needed

Collect your data

Screen your data and make decisions about

- Representativeness
- Missing data
- Univariate Outliers
- Normality

How to report MLR

If creating multiple models, regression results are often best presented in a table.

There is no standard approach– follow that for your domain.

You should at least present the unstandardized or standardized slope (beta), whichever is more interpretable given the data, along with the t-test and the corresponding significance level.

Tables are useful if you find that a paragraph has almost as many numbers as words.

If you do use a table, do not also report the same information in the text. It's either one or the other.